PHYS101&PHYS103/Spring 2013-2014 Midterm 1
SOLUTIONS

PHYSICS 101 (103) NO: _____

NAME: _____

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

- This exam is for 100 minutes and has four problems (Check this now carefully.)
- This is a closed-book/notes/calculator/laptop exam.
- Read the questions carefully and write down your solutions in the space provided neatly and clearly. Write down your final results in the boxes at the end of each question.
- Answers without work shown will not be given any credit.
- A formula sheet is provided for you.

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Problem 1	Problem 2	Problem 3	Problem 4	Total

Problem (1) (25 points)

In an air table experiment a student recorded the position vs time data of a puck which is moving with constant acceleration.

x (m)	y (m)	t (s)
0.200	0	0
0.260	0.008	0.2
0.320	0.032	0.4
0.380	0.072	0.6
0.440	0.128	0.8
0.500	0.200	1.0

(a) (10pts.) Find the acceleration of the puck in both axes and write the acceleration vector in unit vector notation.

One can analyze the two components of the motion separately. In x axes, it is obvious that the particle moves with constant velocity, because the displacements between equal time intervals are the same, so: $\boxed{a_x = 0}$. To find acceleration in y axes one can use $\boxed{y(t) = y(0) + v_y(0)t + at^2/2}$ for any two of the data points. Since it is given that $\boxed{y_0 = 0}$, choosing $\boxed{t = 1.0\ s}$ and $\boxed{t = 0.2\ s}$ will give: $\boxed{0.2 = v_y(0)1 + 0.5a}$ and $\boxed{0.008 = v_y(0)0.2 + 0.5 \times 0.2^2 a}$. Solving the previous two equations one can find $\boxed{v_y(0) = 0}$ and $\boxed{a_y = 0.4\ m/s^2}$.

Thus: $\boxed{\mathbf{a} = 0.4\hat{\mathbf{j}}\ m/s^2}$

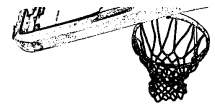
(b) (10pts.) Calculate the position vector of the puck as a function of time in unit vector notation.

For y axes the position of the particle wrt time is simply: $\boxed{\mathrm{y(t) = at^2/2 = 0.2\ m/s^2 \times t^2}}$. For x axes it is simply $\boxed{\mathrm{x(t) = x_0 + v_x \times t}}$. Using x vs t data, one can easily find that $\boxed{\mathrm{x_0 = 0.2\ m}}$ and $\boxed{\mathrm{v_x = 0.3\ m/s}}$

Thus: $\boxed{\mathbf{r}(t) = (0.2m + 0.3m/s \times t)\hat{\mathbf{i}} + 0.2m/s^2 \times t^2\hat{\mathbf{j}}}$

(c) (5pts.) What is the average velocity of the puck between 0.4 and 0.8 seconds? Write it in unit vector notation.

$$\boxed{\mathbf{v}_{\mathrm{av}(0.4-0.8)} = \frac{\mathrm{x}(0.8) - \mathrm{x}(0.4)}{0.8 - 0.4}\hat{\mathbf{i}} + \frac{\mathrm{y}(0.8) - \mathrm{y}(0.4)}{0.8 - 0.4}\hat{\mathbf{j}} = (0.3\hat{\mathbf{i}} + 0.24\hat{\mathbf{j}})\ \mathrm{m/s}}$$



Problem (2) (20 points)

Michael Jordan (1.98 meters tall) is said to have one of the longest hang times (time spent in air) in basketball, almost 1 second.
(You may take g= 10 m/s^2)

(a) (10pts.) What should be his initial vertical velocity for such one-second-hung jump?

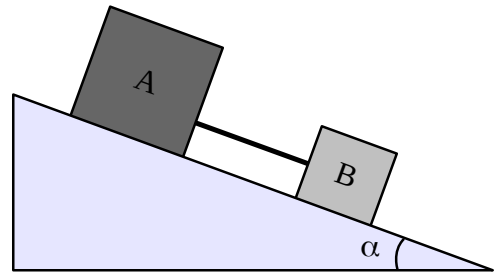
Since time of flight is 1second, he will be at the highest point in 0.5 seconds and $\mathbf{g} = -10$ m/s$^2\hat{\mathbf{j}}$. Vertical component of his velocity at the highest point is ZERO.
Thus: $v_F = v_0 - gt \rightarrow 0 = v_y(0) - 10 \times 0.5 \rightarrow \boxed{\mathrm{v_y}(0) = 5 \text{ m/s}}$.

(b) (10pts.) How high will his head be above the rim level which is 3.05 meters above the ground?

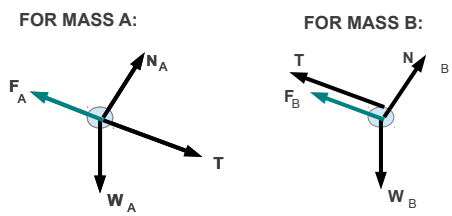
When he is at the highest point, his feet are $y(0.5) = 0 + v_0 t - g(0.5)^2/2 = 1.25\ m$ above the ground. So his head is 1.98 m + 1.25 m = 3.23 m above the ground. Thus his head is 3.23 m − 3.05 m = $\boxed{0.18 \text{ m}}$ above the ground.

Problem (3) (25 points)

Two blocks A and B of masses m_A and m_B respectively are connected by a massless string and slide down an inclined plane which makes α with horizontal axes. The coefficient of kinetic friction between block A and the plane is μ_{A}, and that between block B and the plane is μ_{B}.



(a) (6pts.) Draw the free body diagrams for each of the blocks.



- $\mathrm{F_A} = \mathrm{N_A} \times \mu_{\mathrm{A}}$, the friction force acting on A.
- $\mathrm{N_A} = \mathrm{W_A} \times \cos\alpha$, the normal force acting on A.
- $W_A = m_A \times g$, weight of A.
- $\mathrm{F_B} = \mathrm{N_B} \times \mu_{\mathrm{B}}$, the friction force acting on B.
- $\mathrm{N_B} = \mathrm{W_B} \times \cos\alpha$, the normal force acting on B.
- $W_B = m_B \times g$, weight of B.

(b) (10pts.) If the blocks are moving down with **constant velocity**, find the angle of incline α in terms of $\mathrm{m_A}$, $\mathrm{m_B}$, μ_{A} and μ_{B} .
If we write the Newtons second law for each blocks for the axes paralel to the surface of the incline:

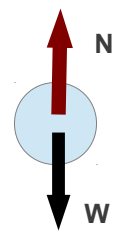
- For A: $\mathrm{T} + \mathrm{W_A}\sin\alpha - \mathrm{F_A} = 0$
  $\rightarrow T = F_A - W_A \sin\alpha = N_A \times \mu_A - m_A g \sin\alpha = \boxed{\mathrm{m_A g}(\cos\alpha\mu_{\mathrm{A}} - \sin\alpha)}$
- For B: $-\mathrm{T} + \mathrm{W_B}\sin\alpha - \mathrm{F_B} = 0$
  $\rightarrow T = -F_B + W_B \sin\alpha = -N_B \times \mu_B + m_B g \sin\alpha = \boxed{\mathrm{m_B g}(-\cos\alpha\mu_{\mathrm{B}} + \sin\alpha)}$
- Since the tensions are equal, $\mathrm{T} = \mathrm{m_A g}(\cos\alpha\mu_{\mathrm{A}} - \sin\alpha) = \mathrm{m_B g}(-\cos\alpha\mu_{\mathrm{B}} + \sin\alpha)$
- solving for α: $\tan\alpha = \frac{\mathrm{m_A}\mu_{\mathrm{A}} + \mathrm{m_B}\mu_{\mathrm{B}}}{\mathrm{m_A} + \mathrm{m_B}} \rightarrow \boxed{\alpha = \arctan\left(\frac{\mathrm{m_A}\mu_{\mathrm{A}} + \mathrm{m_B}\mu_{\mathrm{B}}}{\mathrm{m_A} + \mathrm{m_B}}\right)}$

(c) (9pts.) Calculate the tension of the string if the masses are $\mathrm{m_A} = 80$ kg and $\mathrm{m_B} = 40$ kg, the coefficients of friction are $\mu_{\mathrm{A}} = 0.50$ and $\mu_{\mathrm{B}} = 0.25$, and the tangent of the angle of the incline is $\tan\alpha = 5/12$. Since $\tan\alpha = 5/12$, $\sin\alpha = 5/13$ and $\cos\alpha = 12/13$. Installing all of the given numbers to the one of the equations found for tension above it can be easily found that $\boxed{\mathrm{T} = 800/13 \text{ N}}$.

Problem (4) (30 points)

Assume that you stand on a bathroom scale while riding in an elevator. Will you read a greater or smaller value than your actual mass,

The mass value that the bathroom scale shows is always equal to the normal force applied by the surface divided by g. In any of the cases the free body diagram will contain the weight (W) and the normal force (N) applied by the surface. It is shown at the right side. We have to apply the Newtons second law to find the normal force. In vector language Newtons second law states that $\mathbf{N} - \mathbf{W} = \mathrm{m}\mathbf{a}$.



(a) (10pts.) if the elevator is ascending (moving up) with increasing speed?

Since it has an acceleration upwards, $\mathbf{a} = \mathrm{a}\,\hat{\mathbf{j}}$.
Thus $\mathrm{N} - \mathrm{W} = \mathrm{ma} > 0 \rightarrow \boxed{\mathrm{N} = \mathrm{W} + \mathrm{ma} > \mathrm{W}}$. You weigh more. You read a greater value.

(b) (10pts.) if the elevator is descending (moving down) with constant speed?

Since it has an acceleration zero, $\mathbf{a} = 0$.
Thus $\mathrm{N} - \mathrm{W} = \mathrm{ma} = 0 \rightarrow \boxed{\mathrm{N} = \mathrm{W}}$. You weigh the same. You read the same value.

(c) (10pts.) if the elevator is descending (moving down) with increasing speed?

Since it has an acceleration downwards, $\mathbf{a} = -\mathrm{a}\,\hat{\mathbf{j}}$.
Thus $\mathrm{N} - \mathrm{W} = -\mathrm{ma} < 0 \rightarrow \boxed{\mathrm{N} = \mathrm{W} - \mathrm{ma} < \mathrm{W}}$. You weigh less. You read a smaller value.

(You should give explanations with appropriate free body diagrams.)

Formula Sheet

Scalar Product:

$c = \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

$c = AB \cos\phi$

Vector Product:

$\vec{C} = \vec{A} \times \vec{B}$

$C = AB \sin\phi$

$C_x = A_y B_z - A_z B_y$

$C_y = A_z B_x - A_x B_z$

$C_z = A_x B_y - A_y B_x$

Straight-line motion:

$$v_{av-x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}}$$

$$v_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}} = \frac{dx}{dt}$$

$$a_{av-x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} = \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}}$$

$$a_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}} = \frac{dv_x}{dt}$$

Constant x-acceleration:

$v_x = v_{0x} + a_x t$

$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$

$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$

$$x - x_0 = \left(\frac{v_{0x} + v_x}{2}\right) t$$

Motion in three dimensions:

$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}}$$

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}} = \frac{d\vec{r}}{dt}$$

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}}$$

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}} = \frac{d\vec{v}}{dt}$$

Circular motion:

$$a_{\text{rad}} = \frac{v^2}{R} \quad a_{\text{rad}} = \frac{4\pi^2 R}{T^2}$$

Relative velocity:

$v_{\text{P/A-}x} = v_{\text{P/B-}x} + v_{\text{B/A-}x}$

$\vec{v}_{\text{P/A}} = \vec{v}_{\text{P/B}} + \vec{v}_{\text{B/A}}$

Newton's 1st Law:

$\vec{R} = \vec{F}_1 + \vec{F}_2 + \ldots = \sum \vec{F}$

$\sum \vec{F} = 0$

Newton's 2nd Law:

$\sum \vec{F} = m\vec{a}$

Newton's 3rd Law:

$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}}$

Friction Force:

$f_k = \mu_k n \quad f_s \leq \mu_s n$

Work done by a force:

$W = \vec{F} \cdot \vec{s} = Fs \cos\phi$

Kinetic energy:

$K = \frac{1}{2} m v^2$

The work-energy theorem:

$W_{\text{tot}} = K_2 - K1 = \Delta K$

Work done by a varying force:

$$W = \int_{x_1}^{x_2} F_x \, dx$$

Power:

$$P_{\text{av}} = \frac{\Delta W}{\Delta t}$$

$P_{\text{av}} = \vec{F} \cdot \vec{v}$

Note: If you detach this page during the exam, return it at the end with your exam sheet after you have signed below.

Signature: